

2	(a)	(Net) upward force by a fluid on an object (partially or fully) immersed in the fluid.		1
	(b)	(i)	$M = \rho V = \rho \times \frac{4}{3} (\pi r^3)$ $= 1.21 \times \frac{4}{3} (\pi) (11.0 \times 10^{-2})^3$ $= 0.00675 \text{ kg} = 6.75 \text{ g}$	1 1
		(ii)	 Upthrust, $U = 6.74 \times 10^{-3} \text{ g}$ Force by scale on balloon, $N = 1.55 \times 10^{-3} \text{ g}$ Wt of balloon and air, $W = 1.30 \times 10^{-3} \text{ g} + m_{\text{air}}g$ Net force = 0 Thus $1.30 \times 10^{-3} \text{ g} + m_{\text{air}}g = 6.74 \times 10^{-3} \text{ g} + 1.55 \times 10^{-3} \text{ g}$ Mass of air in balloon, $m_{\text{air}} = 6.99 \text{ g}$	1 1
		(iii)	The air is being compressed by the elastic balloon material so is at a higher pressure and hence denser.	1
	(c)	(i)	At start assume velocity is momentarily zero so drag is zero  Upthrust = $6.75 \times 10^{-3} \text{ g}$ $2.10 \times 10^{-3} \text{ g}$ Wt of balloon and air = $8.30 \times 10^{-3} \text{ g}$ $a = F_{\text{net}}/m$ $= [(8.30 + 2.10 - 6.75) \times 10^{-3} \times 9.81] / [(7.00 + 1.30 + 2.10) \times 10^{-3}]$ $= 3.44 \text{ m s}^{-2}$	1 1
		(ii)	As speed increases, drag force (upward) increases.	1
			Acceleration becomes zero when upthrust + drag force = weight of balloon, air and nut.	1